

Blind vs. Satellite-Detect Uplink in LEO Direct-to-Satellite IoT: An Analytical Threshold

Juan A. Fraire, *Senior Member, IEEE*, Benoit Coeugnet, Oana Iova, André-Luc Beylot, and Vincent Deslandes

Abstract—Direct-to-Satellite IoT (DtS-IoT) networks deploy Low Earth Orbit (LEO) nanosatellites as fast-moving gateways for stationary, mobile, and vehicular sensor platforms. Their intermittent contact windows force a fundamental uplink design choice: transmit *blindly* and accept a non-zero loss probability, or first *detect* an overhead satellite via a downlink beacon. We derive a first-order analytical design rule for this trade-off and obtain a closed-form crossover threshold $N^* = \log \alpha / \log(1 - f_1)$, where $\alpha = \kappa_{\text{eff}} T_b / (2\tau_{\text{tx}})$ is a dimensionless beacon-overhead parameter and f_1 is the single-satellite coverage fraction. Below N^* satellite-detect is more energy-efficient; above it, blind uplink wins. Walker Star propagation simulations validate the model, a parametric study quantifies how altitude, elevation mask, and duty cycle shift N^* , and a lightweight ALOHA collision extension cross-checked by Monte Carlo assesses its sensitivity to offered load. For baseline LoRa-over-satellite parameters ($h=600$ km, $D=1\%$) the threshold is $N^* \approx 220$; for 3GPP NB-IoT NTN under a chipset-averaged $\kappa \approx 0.3$ derived from public module datasheets it shifts to $N^* \approx 546$, identifying the constellation size beyond which a hypothetical lightweight-access mode would outperform today's acquisition-mandatory protocol.

Index Terms—Direct-to-Satellite IoT, LoRa, LoRaWAN, NB-IoT, IoT-NTN, LEO constellations, Walker constellation, energy efficiency, uplink protocol selection.

I. INTRODUCTION

Direct-to-Satellite IoT (DtS-IoT) networks deploy Low Earth Orbit (LEO) nanosatellites as fast-moving gateways for battery-constrained sensors on stationary, mobile, and vehicular platforms that lack terrestrial connectivity [1], [2]. Typical use cases include asset and fleet tracking, container and rolling-stock telemetry, environmental and infrastructure sensing across wide areas, and emergency-alert services from remote or moving deployments. Two complementary technology families are driving this growth, both leveraging commodity chipsets already deployed in billions of terrestrial nodes. The LoRa/LoRaWAN ecosystem builds on ubiquitous LPWAN chipsets, motivating satellite-compatible variants such as LR-FHSS [3] and standard LoRa DtS-IoT [4]. Similarly, 3GPP standardised NB-IoT over non-terrestrial networks (IoT-NTN)

in Release 17 [5], [6], extending the massive installed base of cellular-IoT chipsets to LEO satellite gateways [7], [8].

A satellite in a typical LEO orbit at 500–1200 km altitude moves at roughly 7.5 km/s, creating contact windows of only 5–15 minutes per orbit pass. For a small constellation, the fraction of time that any given device can communicate with a satellite—the *coverage fraction* $f(N)$ —may be well below 50%, meaning that most uplink transmissions miss the satellite entirely. This coverage intermittency motivates two fundamentally different uplink strategies (Fig. 1):

- 1) **Blind uplink:** The device transmits immediately whenever a message is generated, without any satellite awareness. A transmission succeeds with probability $f(N)$; failed transmissions waste the transmit energy E_{tx} . This strategy is realised in practice by LoRaWAN Class A [9].
- 2) **Satellite-detect uplink:** The satellite periodically broadcasts a beacon during its pass. The device duty-cycles its receiver and, upon detecting a beacon, transmits its payload. Success probability approaches unity, but the device spends additional energy $P_{\text{rx}}^{\text{eff}} \mathbb{E}[W]$ listening for the beacon, where $\mathbb{E}[W]$ is the expected waiting time. LoRaWAN Class B and 3GPP NB-IoT over Non-Terrestrial Networks (IoT-NTN) both realise this strategy (see Section III-E).

The trade-off is clear: for a *sparse* constellation ($f(N) \ll 1$), blind transmission wastes most of its energy on missed contact windows, making satellite-detect attractive. Conversely, for a *dense* constellation ($f(N) \rightarrow 1$), blind transmission almost always succeeds while satellite-detect still incurs a non-negligible listening overhead, making blind the better choice. The crossover point N^* —the constellation size at which both protocols have equal energy per delivered message—is the central quantity of this paper.

Prior work has studied blind uplink scalability [10], transmission policies [9], and hybrid Class A/B MAC optimisation [11], but to the best of our knowledge, no analytical model has been proposed to characterise the constellation-size threshold that separates the two protocol regimes.

This paper focuses on *event-driven, coverage-limited* uplink: a single message is generated at a random time and must be delivered through intermittent LEO coverage. We deliberately exclude periodic scheduled reporting (where GNSS-assisted ephemeris wake-up can avoid beacon listening entirely), interference-limited dense-traffic regimes (where a full MAC simulator is needed), and acquisition-heavy protocols such as NB-IoT NTN (where protocol-faithful synchronisation energy accounting is required). Ephemeris-assisted scheduled access is an important alternative for periodic reporting and is

J. A. Fraire is with Inria, INSA Lyon, CITI, UR3720, 69621 Villeurbanne, France; Saarland University, Saarland Informatics Campus, Saarbrücken, Germany; and CONICET-Universidad Nacional de Córdoba, Córdoba, Argentina. E-mail: juan.fraire@inria.fr.

B. Coeugnet is with Kinéis, Toulouse, France; Inria, INSA Lyon, CITI, UR3720, 69621 Villeurbanne, France; and IRIT, ENSEIHT, Toulouse INP, Toulouse, France. E-mail: bcoeugnet@kineis.com.

O. Iova is with INSA Lyon, Inria, CITI, UR3720, 69621 Villeurbanne, France. E-mail: oana.iova@insa-lyon.fr.

A.-L. Beylot is with IRIT, Toulouse, France. E-mail: andre-luc.beylot@toulouse-inp.fr.

V. Deslandes is with Kinéis, Toulouse, France. E-mail: vdeslandes@kineis.com.

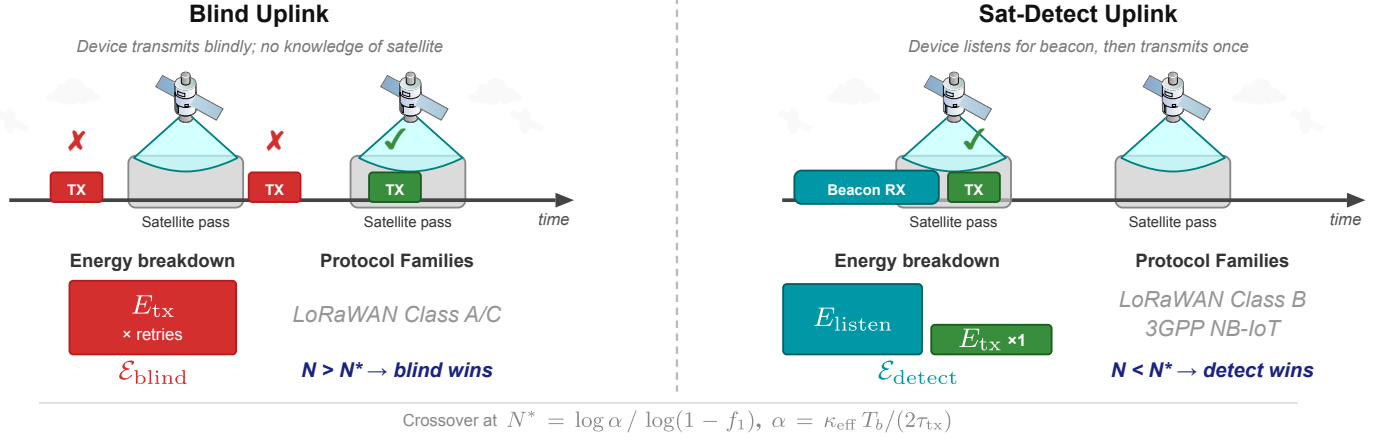


Fig. 1. Conceptual comparison of the two uplink strategies. *Blind* (left): the device transmits at arbitrary times; only the third attempt falls within a satellite pass ($\checkmark$); the first two miss and waste E_{tx} each ($\times$). *Satellite-detect* (right): the device listens for a beacon, then transmits once per detected pass; every transmission succeeds at the cost of a listening overhead E_{listen} . The crossover N^* (Theorem 1) marks the constellation size at which both protocols achieve equal energy per delivered message.

deferred to future work; the present threshold is intended as a compact first-order design rule in the coverage-limited regime.

This paper makes the following contributions:

- An analytical energy-efficiency model for both protocols, expressed in terms of orbital and device parameters (Section III).
- A closed-form expression for N^* (Theorem 1), applicable under the independent-coverage approximation and validated here against selected Walker Star geometries; sparse Walker regimes with bursty pass gaps require propagated or calibrated waiting-time estimates.
- A Monte Carlo simulation campaign using Walker Star propagation to validate the coverage and waiting-time models, including identification of a Poisson-model failure regime at sparse Walker constellations (Section V).
- A parametric sensitivity analysis of N^* with respect to altitude, elevation mask, and receiver duty cycle (Section V).
- A lightweight ALOHA collision/retransmission study showing how offered load shifts N^* , validated by a matching Monte Carlo collision check (Section V-F).
- A qualitative technology mapping of both protocols onto LoRaWAN Class A/B/C and 3GPP IoT-NTN (Section III and V).

II. BACKGROUND AND RELATED WORK

A. Direct-to-Satellite LoRa IoT

DtS-IoT using LoRa has attracted increasing attention as a means of providing low-cost global connectivity for remote sensors [4]. Álvarez et al. [9] study uplink transmission policies in LoRa DtS-IoT and quantify delivery rates for different access strategies, though without comparing beacon-based detection. Herrería-Alonso et al. [10] analyse scalability of the uplink channel for direct-to-satellite LoRa networks, focusing on collision avoidance and Doppler compensation [12] rather than the energy cost of the listen-before-transmit decision. Constellation simulation frameworks such as FLoRaSat [13]

enable end-to-end evaluation but do not expose analytical thresholds for protocol selection as a function of constellation size. Tao et al. [11] propose a hybrid Class A/B MAC protocol for LoRa LEO satellite IoT that optimises the energy-delay trade-off via Markov-chain modelling of phase occupancy, ALOHA collision probability, retransmission/backoff, and fading-aware demodulation success. Their framework captures the full terminal energy budget (sleep, send, receive states) and applies multi-objective optimisation (NSGA-II) to select the Class A/B duty-cycle split. The present paper intentionally keeps a simpler analytical structure—isolating the coverage/listening trade-off in closed form—while borrowing from their collision model the insight that retransmissions multiply energy asymmetrically across protocols (Section V-F).

B. NB-IoT over Non-Terrestrial Networks (IoT-NTN)

IoT-NTN extends NB-IoT connectivity to LEO satellite gateways using the standard 3GPP protocol stack, augmented with NTN-specific system information blocks (SIB31 for satellite ephemeris and Timing Advance, SIB32 for coverage-window prediction) [6]. Kodheli et al. [7] study uplink resource allocation for NB-IoT over LEO, showing that beam partitioning is essential to maintain Doppler offsets within the NB-IoT subcarrier spacing. Sciddurlo et al. [8] provide a full system-level design for NB-IoT over a small LEO constellation, exposing the access latency introduced by the mandatory cell-acquisition chain over sparse coverage. Le et al. [5] survey random-access protocols across both LoRa and NB-IoT in direct-access LEO satellite IoT, characterising the distinct energy trade-offs of each access paradigm. None of these works derives an analytical threshold separating the detect and blind regimes as a function of constellation size, which is the focus of the present paper. Table I summarises the five main access approaches and their treatment herein.

TABLE I
COMPARISON OF UPLINK ACCESS APPROACHES FOR LEO SATELLITE IOT

Approach	Sat. awareness	Main energy cost	Delay	In this paper
Blind / Class A	None	Wasted TX outside coverage	Low attempt	Modelled
Satellite-detect	Beacon listen	RX listening until pass	Waits for coverage	Modelled
Ephemeris-assisted	Predicted passes	GNSS/time sync, guard windows	Periodic, predictable	Future work
Hybrid MAC [11]	A/B Service-adaptive	RX windows + retx	Optimised trade-off	Cited benchmark
NB-IoT NTN	Mandatory acq.	Syn & SIB chain	Standard-controlled	Qualitative only

C. Walker Constellations

A Walker constellation [14] is parameterised as $T/P/F$, where T is the total satellite count, P the number of orbital planes, and F the phasing factor. Walker Delta constellations distribute RAAN planes over 360° and are used when partial coverage at moderate inclinations is acceptable (e.g. 53°). Walker Star constellations distribute RAAN planes over 180° with near-polar inclinations ($\approx 86\text{--}90^\circ$), achieving global coverage including the polar regions. The notation is standardised in [15]. For our global uniform device assumption, Walker Star constellations are the primary reference in simulations.

D. Coverage Models

Coverage fraction for circular orbits over a uniform Earth distribution is well-approximated by the spherical-cap formula $f_1 = (1 - \cos \eta)/2$, where η is the Earth central half-angle subtended by the satellite visibility cone [16]. For N satellites, the independent-satellite model $f(N) = 1 - (1 - f_1)^N$ provides a tractable analytical approximation [17] and typically underestimates the coverage of optimised Walker constellations due to their structured plane phasing.

III. SYSTEM MODEL AND PROTOCOL DEFINITIONS

Table II summarises the parameters and variables used throughout the model.

A. Orbital Model

We consider a Walker constellation of N satellites at altitude h (km) with minimum elevation mask $\varepsilon_{\min}$. Each satellite follows a circular orbit with period

$$T_{\text{orbit}} = 2\pi \sqrt{(R_E + h)^3 / \mu_E}, \quad (1)$$

where $R_E = 6378$ km and $\mu_E = 3.986 \times 10^5 \text{ km}^3 \text{ s}^{-2}$. The Earth central half-angle of the coverage footprint is

$$\eta = \arccos\left(\frac{R_E \cos \varepsilon_{\min}}{R_E + h}\right) - \varepsilon_{\min}, \quad (2)$$

giving a single-satellite instantaneous coverage fraction (fraction of Earth's surface visible):

$$f_1 = \frac{1 - \cos \eta}{2}. \quad (3)$$

TABLE II
MODEL PARAMETERS AND VARIABLES

Symbol	Description	Baseline / range
<i>Orbital / geometric</i>		
N	Constellation size (no. of satellites)	—
h	Orbital altitude	600; 1200 km
$\varepsilon_{\min}$	Minimum elevation mask	$\{10^\circ, 20^\circ, 30^\circ\}$
R_E	Earth radius	6378 km
T_{orbit}	Orbital period (1)	5801; 6565 s
η	Earth central half-angle (2)	—
f_1	Single-satellite coverage fraction (3)	0.45–4.33%
$f(N)$	Constellation coverage fraction (4)	$0 < f(N) \leq 1$
<i>Device / protocol</i>		
τ_{tx}	Uplink frame duration	1 s
T_b	Satellite beacon period (during pass)	30 s
$D = T_{\text{rx}}/T_b$	Receiver duty cycle	10^{-2} (swept $10^{-4}\text{--}1$)
P_{tx}	Transmit power	—
P_{rx}	Receiver active power	—
$\kappa = P_{\text{rx}}/P_{\text{tx}}$	Receive/transmit power ratio	0.1
<i>Derived model quantities</i>		
$E_{\text{tx}} = P_{\text{tx}}\tau_{\text{tx}}$	Transmit energy per frame	—
$P_{\text{rx}}^{\text{eff}} = P_{\text{rx}}D$	Duty-cycled receive power	—
$\kappa_{\text{eff}} = \kappa D$	Effective power ratio	10^{-3} (baseline)
$\mathbb{E}[W]$	Expected wait for first beacon (7)	—
$\mathcal{E}_{\text{blind}}$	Energy per delivered message, blind (5)	—
$\mathcal{E}_{\text{detect}}$	Energy per delivered message, detect (6)	—
$\alpha = \kappa_{\text{eff}}T_b/(2\tau_{\text{tx}})$	Beacon overhead (8)	0.015 (baseline)
N^*	Crossover threshold (10)	≈ 220 (baseline)

For N independent, uniformly distributed satellites, the probability that at least one satellite is visible from a uniformly random device location at a uniformly random time is:

$$f(N) = 1 - (1 - f_1)^N. \quad (4)$$

We adopt (4) as the analytical coverage model; Section V confirms its accuracy against a propagated Walker Star constellation at 86.4° inclination in the constellation-size regime of interest.

B. Protocol Models

1) *Blind Uplink*: A device with a pending message transmits immediately, without knowledge of satellite visibility. The transmission is received iff a satellite is visible at the moment of transmission, which occurs with probability $f(N)$.

Energy per delivered message (blind):

$$\mathcal{E}_{\text{blind}} = \frac{E_{\text{tx}}}{f(N)}, \quad (5)$$

where $E_{\text{tx}} = P_{\text{tx}}\tau_{\text{tx}}$ is the transmit energy, P_{tx} is the transmit power, and τ_{tx} is the uplink frame duration. The baseline value $\tau_{\text{tx}} = 1$ s is representative of a 10-byte LoRaWAN packet at spreading factor SF12, the typical worst-case air-time for LoRa-over-satellite uplinks.

2) *Satellite-Detect Uplink*: During each pass, the satellite broadcasts a periodic beacon of period T_b . The device opens a short receive window of duration T_{rx} every T_b seconds. Upon detecting a beacon, the device transmits its payload in the subsequent uplink opportunity within the same pass.

We define the *effective receive power* $P_{rx}^{\text{eff}} = P_{rx} D$, where P_{rx} is the receiver active power and $D = T_{rx}/T_b$ is the duty cycle.¹ When the device has a message, it incurs a listening overhead $P_{rx}^{\text{eff}} W$ before each successful delivery, where W is the waiting time from message generation to first beacon detected.

Energy per delivered message (satellite-detect):

$$\mathcal{E}_{\text{detect}} = E_{tx} + P_{rx}^{\text{eff}} \mathbb{E}[W]. \quad (6)$$

We adopt a binary line-of-sight channel model: a link succeeds if and only if a satellite is above the elevation mask $\varepsilon_{\min}$, and concurrent uplink collisions are not modelled in the baseline. We further assume the device interacts with at most one satellite at any instant; accordingly, $f(N)$ in (4) denotes the *any-satellite-visible* probability and does not model multi-satellite (superposed) coverage. This isolates the coverage/listening trade-off; collision and retransmission effects are protocol-dependent and are addressed via a lightweight ALOHA extension in Section V-F, while full MAC-layer optimisation is left to specialised models such as [10], [11]. Under the baseline assumptions, satellite-detect delivers every message on the first attempt and no retransmissions are needed. The baseline power ratio $\kappa = P_{rx}/P_{tx} = 0.1$ is representative of LoRa transceivers such as the SX1276 in Class 5 high-power operation ($I_{RX} \approx 12$ mA, $I_{TX} \approx 120$ mA at +20 dBm on the PA_BOOST pin) [18].

Band and regulatory context: The threshold is band-agnostic: it depends only on coverage geometry, beacon timing, and the chipset RX/TX power ratio. LoRa-over-satellite deployments span both sub-GHz ISM (868/915 MHz) and licensed VHF allocations (e.g., Kinéis at $\approx 401/466$ MHz), and the cited chipset values apply across both. The +20 dBm SX1276 operating point used here matches US 915 MHz ISM under frequency hopping and licensed-band operation; the EU 868 MHz +14 dBm EIRP cap would push κ toward 0.3–0.4, shifting N^* upward without altering the structure of the result. The 1% EU868 transmit-time regulation is automatically respected under the event-driven scope adopted throughout this paper.

C. Expected Waiting Time

We model satellite pass arrivals at any ground point as a Poisson process with rate $\nu = N/T_{\text{orbit}}$ and mean pass duration $\bar{\tau}_{\text{pass}} = f_1 T_{\text{orbit}}$, a common analytical approximation for randomised satellite visibility [19]. This Poisson approximation is consistent with the coverage fraction (4) in the linear regime ($Nf_1 \ll 1$) and approaches $f(N)$ for arbitrary N via an exponential gap model; the regime where the approximation breaks down is identified in Section V.

¹In LoRaWAN parlance, “duty cycle” commonly refers to the regulatory transmit-time fraction (e.g. 1% in the EU868 band). Here, instead, D denotes a *receiver* duty cycle: the fraction of T_b during which the device opens a receive window for beacon detection.

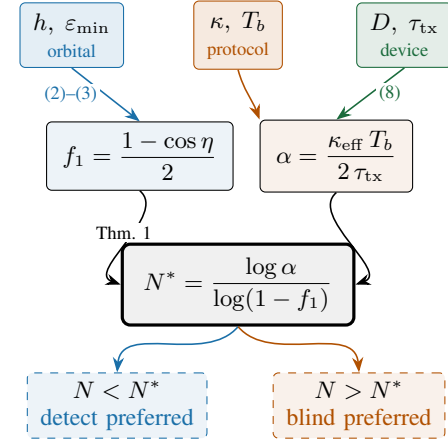


Fig. 2. Parameter influence diagram for the crossover threshold N^* . Orbital/geometric parameters (blue) determine f_1 ; protocol and device parameters (orange/green) determine α . Together they yield N^* (Theorem 1), partitioning the protocol-selection space into the satellite-detect-preferred ($N < N^*$, blue) and blind-preferred ($N > N^*$, orange) regimes.

A device generates a message at a uniformly random time. With probability $f(N)$ it is already in a coverage window, and the expected wait for the next beacon within the pass is $T_b/2$. Otherwise (with probability $1 - f(N)$), it lies in an inter-pass gap; by the memoryless property of the Poisson process, the expected remaining gap duration is $(1 - f(N)) T_{\text{orbit}}/N$. Combining the two cases:

$$\mathbb{E}[W] = \frac{T_b}{2} + (1 - f(N))^2 \frac{T_{\text{orbit}}}{N}. \quad (7)$$

Fig. 2 illustrates how all model parameters interact to produce N^* .

D. Dimensionless Beacon Overhead

The trade-off between (5) and (6) is governed by a single dimensionless quantity, the *beacon overhead*

$$\alpha = \frac{\kappa_{\text{eff}} T_b}{2 \tau_{tx}}, \quad \kappa_{\text{eff}} = \kappa D = \frac{P_{rx} D}{P_{tx}}, \quad (8)$$

which expresses the in-coverage listening cost per uplink opportunity in units of one transmit energy $E_{tx} = P_{tx} \tau_{tx}$. Section IV shows that the crossover threshold N^* depends on the orbital geometry only through f_1 , and on the device and protocol parameters only through α (cf. Fig. 2). With α and κ_{eff} now defined, the next subsection places the main LPWAN satellite standards on this two-parameter map without forward reference to the threshold derivation.

E. Technology Mapping

The two abstract protocols map directly onto existing LPWAN satellite standards.

LoRaWAN Class A (blind). Class A uses pure ALOHA: the device transmits immediately when a message is generated, with no prior interaction with the satellite [9]. A pass is needed only to receive the frame; the device has no satellite awareness. This maps directly to the blind model: $\mathcal{E}_{\text{blind}} = E_{tx}/f(N)$, where τ_{tx} is the LoRa frame air-time and the penalty of

attempted transmissions outside coverage is encoded in the $1/f(N)$ factor.

LoRaWAN Class B (satellite-detect). Class B devices synchronise their receiver to a periodic network beacon. For satellite IoT the beacon originates from the satellite during its pass, and the device duty-cycles at rate D , opening a receive window of duration $T_{rx} = D T_b$ every T_b seconds [9]. Detection triggers the uplink, directly matching (6): the pre-transmission listening overhead is $P_{rx} D \cdot \mathbb{E}[W]$, with T_b and D as explicit model parameters. Class B ping slots are assumed disabled: we account only for beacon detection and the subsequent uplink, not for the optional downlink data delivery on ping slots. Furthermore, standard Class B provides no mechanism for the device to identify which satellite a beacon originated from, and decoding superposed beacons from multiple visible satellites would require sub-microsecond timing jitter across the constellation; both effects are treated as out of scope under our single-satellite-at-a-time assumption.

LoRaWAN Class C (always blind-preferred). Class C keeps the receiver open continuously ($D \rightarrow 1$), yielding $\alpha \approx 1.5$ at baseline LoRa parameters—well above unity. As shown in Section IV, $\alpha \geq 1$ means blind uplink is always the energy-efficient choice regardless of constellation size; there is no N^* .

NB-IoT Non-Terrestrial Networks (IoT-NTN, detect). 3GPP Release 17 standardised NB-IoT over non-terrestrial networks [6], in which a LEO satellite operates as an enhanced node B. An NB-IoT device *must* complete a downlink synchronisation and system-information acquisition chain before transmitting: the Narrowband Primary Synchronisation Signal (NPSS, broadcast every 10 ms), the Secondary Synchronisation Signal (NSSS, every 20 ms), the Master Information Block (every 640 ms), and NTN-specific System Information Block SIB31, which carries satellite ephemeris and a common Timing Advance for propagation-delay pre-compensation. Only after completing this chain can the device transmit an NPRACH random-access preamble and receive an uplink scheduling grant.

Critically, this synchronisation chain is *protocol-mandatory*: there is no mechanism in the current NB-IoT standard to transmit a preamble without prior timing and frequency lock acquired from NPSS/NSSS, making blind uplink impossible by design. Within the satellite-detect framework, the NPSS/NSSS stream acts as the periodic beacon ($T_b \approx 10\text{--}20$ ms) and the active-state RX/TX power split sets κ . Public datasheets of representative NB-IoT chipsets in the +23 dBm class give a range $\kappa \in [0.17, 0.33]$: $\kappa \approx 0.17$ for the Nordic nRF9160 [20] (RX ≈ 46 mA vs. TX ≈ 270 mA), $\kappa \approx 0.23$ for the Quectel BC95-G [21], and $\kappa \approx 0.33$ for the u-blox SARA-N310 [22]. We adopt a representative $\kappa \approx 0.3$ near the upper end of this range, capturing chipsets with less efficient RF front-ends; lower- κ modules would shift N^* further upward. This κ captures the active-state RX/TX power split but not the full NPSS/NSSS/MIB/SIB31 acquisition energy, which depends on signal strength, chipset architecture, and synchronisation time; a protocol-faithful acquisition-energy model is left for future work. Because $T_b \approx 10\text{--}20$ ms is three orders of magnitude shorter than typical LoRa beacon periods, α is

small and the crossover N^* is pushed to larger values than for LoRa; where exactly it falls is quantified in Section V.

Remark 1. Both LoRaWAN Class B and IoT-NTN implement the satellite-detect protocol; they differ in the richness of the acquisition chain (typically much longer beacon periods for Class B vs. ≈ 20 ms for NPSS) and in κ . The model parameters κ_{eff} and T_b are sufficient to map either technology onto the crossover formula (10) at the level of the active-state power split; an acquisition-procedure-faithful model for NB-IoT is left for future work.

IV. ANALYTICAL THRESHOLD N^*

A. Crossover Condition

Using $E_{\text{tx}} = P_{\text{tx}} \tau_{\text{tx}}$ and κ_{eff} as defined in (8), equating $\mathcal{E}_{\text{blind}} = \mathcal{E}_{\text{detect}}$ yields

$$\kappa_{\text{eff}} \frac{\mathbb{E}[W(N)]}{\tau_{\text{tx}}} = \frac{1 - f(N)}{f(N)}. \quad (9)$$

The left-hand side is the normalised listening overhead; the right-hand side is the normalised wasted-transmission cost. Satellite-detect is preferable when the listening overhead is below the wasted-transmission cost ($N < N^*$), whereas blind transmission becomes preferable when it exceeds it ($N > N^*$).

B. Closed-Form Threshold

Theorem 1 (Crossover threshold). *In the dense-constellation regime ($f(N^*) \rightarrow 1$, i.e. $\alpha \ll 1$), the crossover threshold satisfies*

$$N^* = \frac{\log \alpha}{\log(1 - f_1)}, \quad (10)$$

with α the dimensionless beacon overhead defined in (8).

Derivation. For $f(N^*) = 1 - u^*$ with $u^* \rightarrow 0$, the gap term $(1 - f)^2 T_{\text{orbit}}/N$ in (7) is $O(u^{*2})$ and is dominated by $T_b/2$. Equation (9) reduces to $\kappa_{\text{eff}} T_b / (2\tau_{\text{tx}}) \approx u^* = 1 - f(N^*)$, i.e. $\alpha \approx (1 - f_1)^{N^*}$. Solving for N^* yields (10). $\square$

Remark 2. Equation (10) reveals that N^* depends on only two dimensionless quantities: the beacon overhead α and the single-satellite coverage f_1 . Larger f_1 (higher altitude or wider footprint) reduces N^* because each satellite contributes more coverage, so the dense-coverage regime is reached with fewer satellites. Larger α (more listening overhead per coverage pass) also reduces N^* : blind uplink becomes preferable at smaller constellation sizes because the cost of listening dominates earlier.

C. Tabulated Values and Numerical Refinement

Formula (10) is exact in the limit $\alpha \rightarrow 0$ and provides sub-1% accuracy for $\alpha \lesssim 0.05$, corresponding to $f(N^*) > 0.95$. For larger α , one iteration of Newton's method on the full crossover condition (9) with (7) substituted gives $< 0.5\%$ error. Table III illustrates typical values across four decades of duty cycle; a more complete classification of validity regimes, including the sparse-Walker breakdown, follows in Section IV-D.

TABLE III
CROSSOVER THRESHOLD N^* FOR $h=600$ KM, $\varepsilon_{\min}=10^\circ$, $\kappa=0.1$,
 $\tau_{tx}=1$ s, $T_b=30$ s ($f_1 = 1.89\%$, $T_{\text{orbit}} = 5801$ s).

Duty cycle D	κ_{eff}	α	$f(N^*)$	N^*
1.0	0.100	1.500	—	— [‡]
0.10	0.010	0.150	0.075	— [†]
0.01	0.001	0.015	0.985	220
0.001	0.0001	0.0015	0.999	340
0.0001	0.00001	0.00015	0.9999	460
0.00001	0.000001	0.000015	≈ 1	581

[†] Analytical: two crossovers at $N \approx 4$ and $N \approx 105$ (detect preferred between them); simulation: no crossover—see Section V-E. [‡] $\alpha \geq 1$: blind always preferred, no crossover exists.

TABLE IV
CROSSOVER THRESHOLD N^* AT THE LoRaWAN CLASS B STANDARD
BEACON PERIOD $T_b=128$ s, COMPARED TO THE $T_b=30$ s BASELINE.

D	$T_b = 30$ s (baseline)		$T_b = 128$ s (Class B std.)	
	α	N^*	α	N^*
1.0000	1.5000	— [‡]	6.4000	— [‡]
0.1000	0.1500	— [†]	0.6400	— [†]
0.0100	0.0150	220	0.0640	147*
0.0010	0.0015	340	0.0064	264
0.0001	0.0002	460	0.0006	384

[†] Sparse-Walker artefact (see Table V); [‡] $\alpha \geq 1$: blind always preferred; * moderate regime ($\alpha > 0.05$), numerical solution preferred.

When $\alpha \geq 1$ (the LoRaWAN Class C regime: $D \approx 1$, $\alpha \approx 1.5$), the listening overhead at any coverage level already exceeds the wasted-transmission cost even for a single satellite, and blind uplink is always preferable. Satellite-detect is therefore only beneficial when $\alpha < 1$, i.e. the duty-cycled listening overhead per coverage pass is less than one transmit-energy equivalent.

Standard Class B beacon period: The LoRaWAN Class B specification fixes the network beacon period at $T_b=128$ s, whereas the baseline of Table III uses $T_b=30$ s as a tunable design parameter representative of proprietary or extended Class-B-like satellite implementations. Because $\alpha \propto T_b$, raising T_b from 30 to 128 s scales the beacon-overhead parameter by a factor of $128/30 \approx 4.3$, shifting the crossover threshold downward. Table IV reports the resulting N^* values across the same duty-cycle sweep; in the dense regime ($D \leq 10^{-2}$) the threshold drops by roughly 30–35%, consistent with the logarithmic dependence in (10). For $D = 10^{-1}$, both T_b values fall in the sparse-Walker regime ($\alpha \geq 0.15$, see Table V) where the closed-form roots are model artefacts rather than operative crossovers; the standard $T_b=128$ s case at $D = 10^{-2}$ ($\alpha \approx 0.06$) sits at the edge of the moderate regime and is also flagged accordingly.

D. Validity Regime

Equation (10) is a first-order approximation derived under two assumptions: (i) independent-satellite coverage (4), and (ii) a Poisson gap model for the inter-pass waiting time (7). Both hold tightly in the dense regime but degrade for sparse

TABLE V
VALIDITY REGIMES FOR THE ANALYTICAL THRESHOLD

Regime	Condition	Recommended use
Dense, low overhead	$\alpha \leq 0.05$, $f(N^*) \geq 0.95$	Closed-form N^* (10)
Moderate	$0.05 < \alpha < 0.1$	Numerical solution of (9)
Sparse Walker	Low N , few planes, bursty gaps	Propagated simulation required
High overhead	$\alpha \geq 1$	Blind always preferred; no N^*

TABLE VI
RELATIVE ERROR OF ANALYTICAL $\mathbb{E}[W]$ VS WALKER STAR SIMULATION
(BASELINE $h=600$ KM, $\varepsilon_{\min}=10^\circ$, $T_b=30$ s)

N	$\mathbb{E}[W]_{\text{ana}}$ (s)	$\mathbb{E}[W]_{\text{sim}}$ (s)	Rel. error
4	1259	5020	+298.5%
20	150	632	+321.8%
49	33	98	+196.3%
90	17	27	+55.7%
196	15	15	−0.1%
400	15	15	−0.0%
700	15	15	−0.0%
2000	15	15	+0.0%

Walker configurations with bursty pass geometry. Table V classifies four operating regimes and the recommended analytical tool for each.

Cautionary example. For $D = 0.1$ ($\alpha = 0.15$), the analytical model predicts dual crossovers at $N \approx 4$ and $N \approx 105$. However, Walker Star simulation at $N = 8$ yields $\mathbb{E}[W]_{\text{sim}} \approx 4109$ s versus 540 s analytical—a $7.6\times$ overestimate by the Poisson gap model due to plane-synchronised pass bursts. With realistic waiting times, the detect listening overhead exceeds the blind penalty at all tested sizes, and no operative crossover exists (Section V-E). This confirms that the threshold should be trusted primarily for $\alpha \ll 1$ (i.e. $D \ll 1$), where N^* falls in the dense regime and the Poisson approximation is accurate.

Systematic error quantification. Table VI reports the relative error of the analytical $\mathbb{E}[W]$ against Walker Star simulation across the baseline configuration ($h = 600$ km, $\varepsilon_{\min} = 10^\circ$, near-polar inclination 86.4° , $T_b = 30$ s). The error exceeds +50% for $N \lesssim 90$ (sparse Walker, correlated pass geometry), and drops below 0.1% by $N = 196$. Throughout the relevant LoRa crossover region ($N^* \approx 220$ at $D = 0.01$), the Poisson approximation is accurate to within 1%, confirming that N^* predictions are reliable in the dense regime.

Ground-point sampling confidence interval. To address the concern that the crossover decision $\mathcal{E}_{\text{blind}}/\mathcal{E}_{\text{detect}} = 1$ may be sensitive to statistical sampling, we re-ran the baseline simulation at $N \in \{100, 150, 200, 220, 300\}$ with three different ground-point seeds (seeds 42, 43, 44; $M=300$ points each). Table VII reports the resulting mean and standard deviation of $f(N)$, $\mathbb{E}[W]$, and the efficiency ratio. In the transition region ($N = 100$, $f \approx 0.87$), the 1σ ground-point CI is $\lesssim 0.3\%$ on the efficiency ratio. In the dense regime

TABLE VII

GROUND-POINT SAMPLING CI AT THE BASELINE CROSSOVER REGION:
 $h=600$ km, $\varepsilon_{\min}=10^\circ$, $D=0.01$, $M=300$ GROUND POINTS, 3 SEEDS.

N	$f(N)$	$\mathbb{E}[W]$ (s)	ratio $\mathcal{E}_{\text{blind}}/\mathcal{E}_{\text{detect}}$
100	0.8672 ± 0.0021	25.1 ± 0.4	1.1249 ± 0.0025
150	1.0000 ± 0.0000	15.0 ± 0.0	0.9852 ± 0.0000
200	1.0000 ± 0.0000	15.0 ± 0.0	0.9852 ± 0.0000
220	1.0000 ± 0.0000	15.0 ± 0.0	0.9852 ± 0.0000
300	1.0000 ± 0.0000	15.0 ± 0.0	0.9852 ± 0.0000

($N \geq 150$), coverage saturates to unity and $\mathbb{E}[W]$ hits the $T_b/2$ floor at the simulation resolution ($\Delta t=60$ s), so the efficiency ratio is reproducible to numerical precision across seeds. The crossover prediction $N^* \approx 220$ is therefore insensitive to ground-point sampling noise.

Statistical scope. The CI above captures variability of the simulator under different uniform ground-point samples; it does not capture variability under different Walker phasing factors F , time epochs, or regional traffic distributions. A full statistical characterisation across those dimensions is left for future work; the targeted simulations here are used to identify whether the analytical threshold lies inside or outside its validity regime, not to establish a complete uncertainty envelope.

V. EVALUATION

A. Simulation Setup

We validate the analytical model using a Walker Star constellation at 86.4° inclination propagated via a two-body J_2 propagator. Ground points are sampled uniformly on the sphere ($M = 300$ points); coverage is checked every $\Delta t = 60$ s over a 24-hour window. The constellation is constructed following the Walker $T/P/F$ notation of [15], with P planes chosen as the divisor of N nearest to $\sqrt{N}$ (with $P \geq 2$ to avoid degenerate single-plane configurations), and $F = 0$. Eight constellation sizes spanning $N \in \{4, \dots, 2000\}$ are simulated for the main coverage/waiting-time validation, and three configurations are exercised: the baseline ($h = 600$ km, $\varepsilon_{\min} = 10^\circ$) and the two extreme curves ($h = 1200$ km, $\varepsilon_{\min} = 10^\circ$) and ($h = 600$ km, $\varepsilon_{\min} = 30^\circ$). For each size, the simulated $\mathbb{E}[W]$ is computed as the mean gap to first satellite visibility plus $T_b/2$, consistent with the analytical definition (7). A second, targeted run validates Fig. 6 (Section V-E): for $D = 0.1$, twelve sizes spanning $N \in \{1, 4, 5, 8, 10, 20, 50, 90, 100, 110, 120, 150\}$ confirm the sparse-regime Poisson artefact; for $D \in \{1, 0.01, 0.001, 0.0001\}$, five sizes bracketing each analytical N^* validate the crossover prediction. A third, multi-seed run (seeds 42, 43, 44) supplies the ground-point sampling confidence interval of Table VII. All code and data are publicly available².

Analytical curves in Figs. 3–4 are shown for all combinations of $\varepsilon_{\min} \in \{10^\circ, 20^\circ, 30^\circ\}$ and $h \in \{600, 1200\}$ km;

²<https://gitlab.inria.fr/jfraire/blind-vs-detect> (will be made public in the final camera-ready version if the paper is accepted).

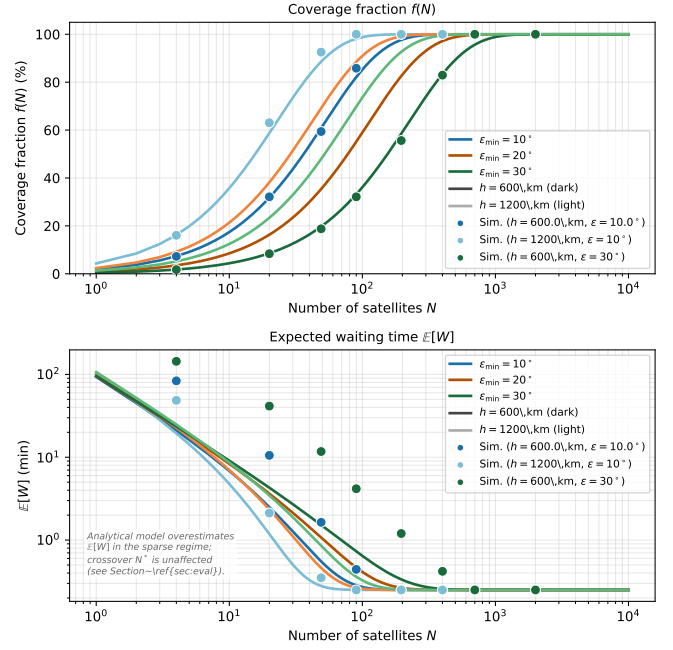


Fig. 3. Coverage fraction $f(N)$ (top) and expected waiting time $\mathbb{E}[W]$ (bottom) vs. constellation size N . Analytical curves (4)–(7) for $\varepsilon_{\min} \in \{10^\circ, 20^\circ, 30^\circ\}$ (blue/orange/green) at $h = 600$ km (dark) and $h = 1200$ km (light). Simulation markers: Walker Star Monte Carlo for three configurations—dark-blue ($h = 600$ km, $\varepsilon_{\min} = 10^\circ$), light-blue ($h = 1200$ km, $\varepsilon_{\min} = 10^\circ$), dark-green ($h = 600$ km, $\varepsilon_{\min} = 30^\circ$); $N = 4$ –2000 in each case. Residual overestimation of $\mathbb{E}[W]$ at low N is due to correlated pass geometry in Walker Star not captured by the Poisson gap model.

simulation markers are provided for three representative configurations: the baseline ($h = 600$ km, $\varepsilon_{\min} = 10^\circ$) and the two extreme curves ($h = 1200$ km, $\varepsilon_{\min} = 10^\circ$) and ($h = 600$ km, $\varepsilon_{\min} = 30^\circ$). Unless otherwise stated: $T_b = 30$ s, $\kappa = 0.1$, $D = 0.01$, $\tau_{\text{tx}} = 1$ s.

B. Coverage Fraction and Waiting Time

Fig. 3 shows $f(N)$ and $\mathbb{E}[W]$ for all six parameter combinations alongside simulation markers for the baseline. The coverage fraction model (4) matches the simulation closely; the slight overperformance of the Walker Star at higher N is expected, as an optimised constellation provides more uniform coverage than the independent-satellite approximation.

As expected, higher altitude ($h = 1200$ km) yields substantially larger $f(N)$ for the same N : the bigger footprint compensates for fewer planes. A stricter elevation mask reduces the footprint and lowers $f(N)$, shifting saturation to a larger N . Both effects are captured accurately by (4) once the corresponding f_1 is inserted.

The simulated waiting time agrees well with (7) for moderate to large N , and converges to $T_b/2 = 15$ s as $N \rightarrow \infty$ regardless of altitude or elevation mask (the gap vanishes and only the in-pass beacon wait remains). At low N , the simulation gives somewhat higher $\mathbb{E}[W]$ than the analytical model: with few orbital planes, Walker Star pass gaps are correlated and occasional long voids exceed what the Poisson

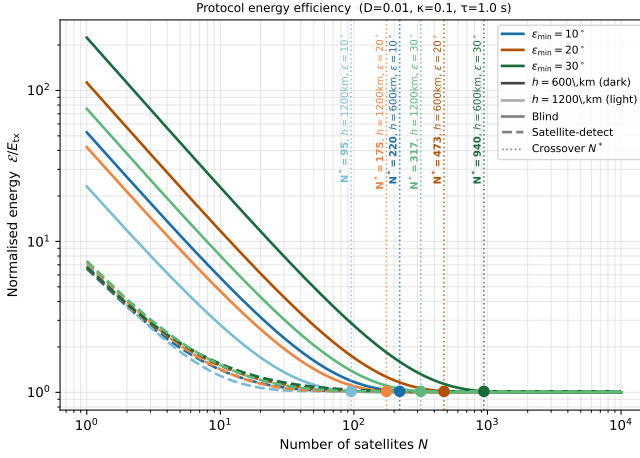


Fig. 4. Normalised energy per delivered message $\mathcal{E}/E_{tx}$ for blind (solid) and satellite-detect (dashed) protocols vs. constellation size N (log-log scale). Curves shown for $\varepsilon_{\min} \in \{10^\circ, 20^\circ, 30^\circ\}$ (blue/orange/green) at $h = 600$ km (dark) and $h = 1200$ km (light). Vertical dotted lines and bold labels mark the crossover threshold N^* for each combination. $D = 0.01$, $\kappa = 0.1$, $\tau_{tx} = 1$ s.

model predicts. This discrepancy is concentrated in the sparse regime ($N \ll N^*$) and does not affect the crossover analysis.

C. Energy per Delivered Message and Crossover

Fig. 4 illustrates the main result for all six combinations of elevation mask and orbital altitude. For sparse constellations ($N \ll N^*$), the blind protocol wastes most of its transmit energy on missed passes; satellite-detect is significantly more energy-efficient. As N increases towards N^* , $\mathcal{E}_{\text{blind}}$ drops rapidly while $\mathcal{E}_{\text{detect}}$ decreases more slowly (the in-coverage beacon overhead $T_b/2$ is always present). At the crossover, both curves intersect; beyond N^* , blind is the better choice.

Effect of elevation mask. A stricter $\varepsilon_{\min}$ reduces f_1 and therefore raises N^* . At $h = 600$ km the threshold rises from $N^* \approx 220$ (10°) to ≈ 475 (20°) and ≈ 930 (30°), roughly doubling with each 10° increment. The 30° case exceeds the size of any current LEO IoT constellation, so satellite-detect remains preferable throughout the foreseeable deployment range under strict link-margin constraints—a regime typical of mobile or vehicular terminals operating without an open sky.

Effect of orbital altitude. Higher altitude enlarges the satellite footprint, yielding a larger f_1 and a lower N^* for the same $\varepsilon_{\min}$. At $h = 1200$ km the thresholds drop to $N^* \approx 94$, 174, and 316 for $\varepsilon_{\min} = 10^\circ$, 20° , and 30° , respectively—a reduction by roughly a factor of 2.5–3 compared to $h = 600$ km. This confirms the altitude sensitivity already evident in Fig. 5(a) and underscores that higher-orbit constellations reach the blind-favoured regime with far fewer satellites.

D. Sensitivity of N^*

Fig. 5 shows how N^* varies with each system parameter; multiple curves per subplot expose the joint influence of the remaining parameters.

Altitude [Fig. 5(a)]. Increasing h from 300 to 2000 km enlarges the satellite footprint (f_1 rises from 0.6% to 12%) while also lengthening T_{orbit} . The net effect is a monotone decrease in N^* : fewer satellites are needed to reach the threshold coverage fraction $f(N^*) \approx 1 - \alpha$. A stricter elevation mask (orange, green) shifts the entire curve upward, since a smaller f_1 requires more satellites to achieve the same coverage level $f(N^*)$. Reducing D from 0.01 to 0.001 (blue dashed) also shifts N^* upward uniformly— α decreases by one order of magnitude, so the required coverage $(1 - \alpha)$ increases—but does not change the shape of the altitude dependence.

Duty cycle [Fig. 5(b)]. D is the parameter with the widest practical tuning range: spanning four decades (10^{-4} to 1), it drives N^* from ≈ 100 to ≈ 580 at baseline before the crossover disappears entirely at $\alpha \geq 1$. Decreasing D reduces α and raises N^* : satellite-detect remains preferable up to larger constellations. The figure overlays two altitudes and two elevation masks. Higher altitude (light shades) lowers N^* at every D , while a stricter mask (orange) raises it, consistent with their opposing effect on f_1 . For $D = 10^{-3}$, $N^* \approx 340$ at baseline, encompassing today's largest operational LEO IoT constellations. For large D (approaching 1), $\alpha \geq 1$ and satellite-detect is never preferable; the crossover disappears. For $D \gtrsim 0.094$ ($\alpha \gtrsim 0.141$), the crossover condition (9) admits two roots: a low- N root (blind→detect transition, dotted curves) and the operative high- N root (detect→blind, solid curves). Both branches are plotted explicitly; the solid branch traces N^* smoothly from ≈ 580 down to ≈ 100 as D increases, while the dotted branch appears at $N \lesssim 10$ and represents a sparse-regime artefact of the Poisson gap model. As confirmed by Monte Carlo simulation (Section V-E), no real crossover exists at such sparse constellations; the lower root therefore carries no operational significance. Both roots merge and vanish as $\alpha \rightarrow 1$ ($D \approx 0.67$), beyond which blind transmission is always preferable regardless of constellation size.

Elevation mask [Fig. 5(c)]. A stricter $\varepsilon_{\min}$ reduces f_1 (smaller footprint) and thus raises N^* . Each additional 10° roughly doubles N^* , consistent with the logarithmic sensitivity $N^* \propto 1/|\log(1 - f_1)|$ in (10) and the values in Section V. The subplot combines two altitudes (dark/light blue) and two duty cycles (solid/dashed): a higher altitude lowers N^* across the board, while a lower D raises it. The two effects are approximately additive on the log scale, confirming that altitude and duty cycle act as independent scaling factors on N^* .

E. Effect of Duty Cycle Across the Constellation-Size Range

Fig. 6 plots the efficiency ratio $\mathcal{E}_{\text{blind}}/\mathcal{E}_{\text{detect}}$ on a log-log scale for five values of D spanning four decades. A ratio above 1 means satellite-detect is more energy-efficient; the dashed line at ratio = 1 marks the crossover N^* . The y-axis is clipped at 10: for ratio $\gg 1$ the satellite-detect advantage is already decisive, and the incremental gain of a larger ratio is not practically relevant.

For $D = 1$ ($\alpha = 1.5 > 1$), blind is the definitive winner: the ratio is below 1 for all N , and no crossover exists. For

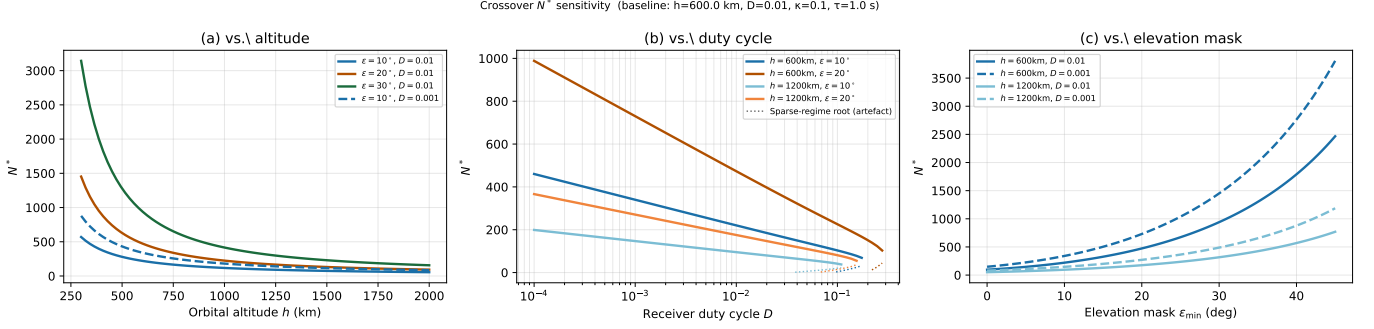


Fig. 5. Crossover threshold N^* as a function of (a) orbital altitude h , (b) receiver duty cycle D , and (c) elevation mask $\varepsilon_{\min}$. Colours follow the same convention as Figs. 3–4: blue/orange/green for $\varepsilon_{\min} = 10^\circ/20^\circ/30^\circ$; dark/light shades for $h = 600/1200$ km; dashed lines for $D = 0.001$. (a) Three elevation masks at $D = 0.01$ (solid) plus $\varepsilon_{\min} = 10^\circ$ at $D = 0.001$ (blue dashed), illustrating the joint altitude–duty-cycle sensitivity. (b) Two altitudes and two elevation masks; solid curves show the operative (high- N) root, dotted curves the sparse-regime artefact root that appears for $D \gtrsim 0.09$. (c) Two altitudes (dark/light blue) and two duty cycles ($D = 0.01$ solid, $D = 0.001$ dashed). Remaining parameters: $\kappa = 0.1$, $\tau_{\text{tx}} = 1$ s, $T_b = 30$ s.

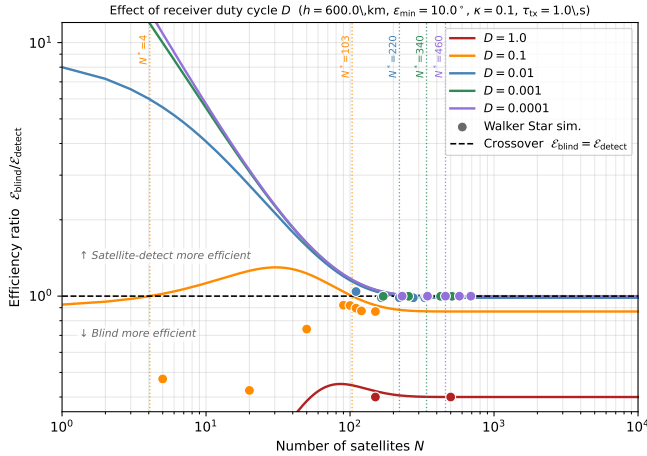


Fig. 6. Efficiency ratio $\mathcal{E}_{\text{blind}}/\mathcal{E}_{\text{detect}}$ vs. constellation size N (log-log scale, $h = 600$ km, $\varepsilon_{\min} = 10^\circ$, $\kappa = 0.1$, $\tau_{\text{tx}} = 1$ s). Ratio > 1 : satellite-detect saves energy; ratio < 1 : blind is preferable. $D = 1$ has no crossover (blind always wins); for $D \leq 0.01$ each analytical curve shows a distinct N^* (vertical dotted lines). Filled circles: Walker Star Monte Carlo simulation, colour-matched to the corresponding analytical curve. For $D = 0.1$ (orange), simulated ratios stay below 1 at all tested sizes, confirming the analytical dual-crossover is a model artefact (see text). For $D \leq 0.01$, simulation points bracket the analytical N^* , validating the crossover prediction in the dense-constellation regime. Y-axis clipped at 10: above this value, satellite-detect is the overwhelming winner, and the exact ratio is not operationally relevant.

$D = 10^{-1}$ ($\alpha = 0.15$), the analytical curve predicts a non-monotonic shape with two crossovers — detect preferred only for $4 \lesssim N \lesssim 105$ — but targeted Monte Carlo simulation (orange circles) tells a strikingly different story: the simulated ratio stays *below 1 at all tested sizes* ($N = 1$ –150), meaning blind is always preferable and no operative crossover exists.

The root cause is the Poisson gap model’s failure at small Walker Star constellations. With few orbital planes (e.g. $P = 2$ for $N = 8$ or $N = 10$), satellites arrive in plane-synchronised bursts separated by long inter-burst gaps, producing actual waiting times 3 – $8\times$ larger than the Poisson prediction (e.g. $\mathbb{E}[W]_{\text{sim}} = 4109$ s vs. 540 s analytical at $N = 8$). With

such elevated $\mathbb{E}[W]$, $\kappa_{\text{eff}} \mathbb{E}[W]$ remains large enough that blind—though it wastes many transmissions—is still cheaper than the listening overhead at $D = 0.1$. This confirms that the analytical crossover model should be trusted primarily for $\alpha \ll 1$ (i.e. $D \ll 1$), where N^* falls in the dense-constellation regime, and the Poisson approximation is accurate. For $D = 10^{-2}$ ($\alpha = 0.015$), the ratio starts just below 8 at $N = 1$ (fully within the plotted range) and crosses unity at $N^* \approx 220$; this is the baseline scenario of Table III. For $D = 10^{-3}$ and $D = 10^{-4}$, the curves enter the plot from above the y-clip and cross unity at $N^* \approx 340$ and ≈ 460 , respectively. All five curves together illustrate the full range of behaviour: from “blind always wins” at $D = 1$ to satellite-detect remaining preferable up to hundreds of satellites at aggressive duty cycles.

Operationally, Fig. 6 thus serves as a quick-lookup chart for constellation designers and terminal integrators: given a target constellation size N and a device duty cycle D , the corresponding curve directly indicates whether blind or satellite-detect uplink is the energy-efficient choice for stationary or mobile DtS-IoT terminals.

F. Collision and Retransmission Sensitivity

The baseline model assumes collision-free transmissions. We now extend it with a lightweight unslotted ALOHA model to assess how offered load shifts the crossover; this is intentionally illustrative, not a complete MAC model [11].

Let $p_{\text{ch}} = 1$ (binary LoS channel) and define per-mode success probabilities under ALOHA load G :

$$p_{\text{col},b} = e^{-2G_b}, \quad p_{\text{col},d} = e^{-2G_d}, \quad (11)$$

$$p_b = f(N) p_{\text{col},b}, \quad p_d = p_{\text{col},d},$$

where G_b and G_d are the offered loads (in Erlang) seen by blind and detect transmissions, respectively. With a retransmission cap R , the expected number of attempts is

$$\mathbb{E}[A(p, R)] = \sum_{r=1}^R r(1-p)^{r-1}p + R(1-p)^R, \quad (12)$$

TABLE VIII
CROSSOVER N^* UNDER ALOHA COLLISION LOAD ($D=0.01$, $R=8$)

G_b	G_d	$p_{\text{col},b}$	$p_{\text{col},d}$	N^*
0	0	1.000	1.000	220
0.05	0.05	0.905	0.905	220
0.10	0.10	0.819	0.819	220
0.10	0.02	0.819	0.961	— [†]

[†] No crossover: asymmetric load penalises blind disproportionately, making detect preferred at all N .

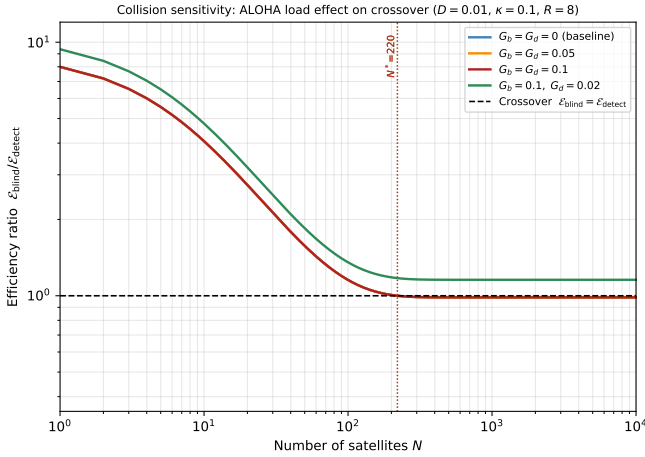


Fig. 7. Efficiency ratio $\mathcal{E}_{\text{blind}}^{\text{col}}/\mathcal{E}_{\text{detect}}^{\text{col}}$ vs. N under ALOHA collision load (log-log, $D=0.01$, $\kappa=0.1$, $R=8$). Symmetric loads preserve the baseline $N^* \approx 220$; asymmetric load ($G_b=0.1$, $G_d=0.02$, green) pushes the ratio above 1 everywhere, eliminating the crossover.

and the delivery probability is $P_{\text{del}}(p, R) = 1 - (1-p)^R$. The extended energy per delivered message becomes

$$\mathcal{E}_{\text{blind}}^{\text{col}} = E_{\text{tx}} \frac{\mathbb{E}[A(p_b, R)]}{P_{\text{del}}(p_b, R)}, \quad (13)$$

$$\mathcal{E}_{\text{detect}}^{\text{col}} = (E_{\text{tx}} + \kappa_{\text{eff}} \mathbb{E}[W]) \frac{\mathbb{E}[A(p_d, R)]}{P_{\text{del}}(p_d, R)}. \quad (14)$$

Table VIII reports the numerical crossover for four load scenarios at baseline parameters with $R = 8$. *Symmetric* loads ($G_b = G_d$) barely shift N^* because both protocols suffer equally from collisions—the threshold is determined by the coverage/listening trade-off, not by collision rate. *Asymmetric* load ($G_b = 0.1$, $G_d = 0.02$)—reflecting that blind transmissions occur at arbitrary times and thus face higher contention than coordinated detect transmissions—eliminates the crossover entirely: the collision-amplified blind penalty makes detect preferred at all constellation sizes. Fig. 7 visualises the efficiency ratio for all four cases.

This extension confirms that the collision-free baseline is representative when both protocols experience similar contention, but that the threshold is sensitive to differential load—a regime in which fuller MAC models [10], [11] are needed.

Monte Carlo sanity check. To verify the lightweight ALOHA extension, Table IX compares analytical and simulated efficiency ratios at three offered loads ($G \in \{0, 0.05, 0.1\}$) for constellation sizes spanning the baseline

TABLE IX
MONTE CARLO SANITY CHECK: ANALYTICAL VS SIMULATED EFFICIENCY RATIO $\mathcal{E}_{\text{blind}}/\mathcal{E}_{\text{detect}}$ UNDER SYMMETRIC ALOHA LOAD ($D=0.01$, $R=8$, 3 SEEDS)

N	$G = 0$		$G = 0.05$		$G = 0.1$	
	ana.	sim.	ana.	sim.	ana.	sim.
50	1.57	2.88	1.57	1.41	1.57	1.10
100	1.15	2.64	1.15	1.72	1.15	1.32
200	1.01	1.28	1.01	1.26	1.01	1.25
300	0.99	0.98	0.99	1.03	0.99	1.06
500	0.99	0.98	0.99	1.00	0.99	1.04

crossover region. The simulator uses Poisson arrivals (rate $\lambda = G/(2\tau_{\text{tx}})$), fixed packet duration τ_{tx} , overlap-based collisions, and a retransmission cap $R = 8$ over a simulation window of 50 000 s with three random seeds per point. Agreement is tight ($\lesssim 5\%$) for $N \geq 300$, the regime relevant to $N^* \approx 220$; deviations grow at $N = 50$ – 100 where the simplified Poisson coverage timeline embedded in the simulator overstates the wait penalty relative to the analytical $\mathbb{E}[W]$. This is a consistency check for the analytical collision extension, not a full MAC-layer performance evaluation.

G. Technology Placement in the Protocol Space

The figures above can now be read as a protocol-selection map for the technologies introduced in Section III-E. *LoRaWAN Class A* (blind) is energy-competitive only above N^* . At the baseline $D = 0.01$, $N^* \approx 220$, which exceeds the size of most current DtS-IoT constellations. Under the collision-free, event-driven, globally uniform baseline, Class B-like detection is more energy-efficient for these constellation sizes; scheduled access, traffic load, and retransmissions can shift this boundary (cf. Section V-F and [11]). *LoRaWAN Class B* ($D = 0.01$, $\kappa = 0.1$) maps precisely onto the baseline $D = 0.01$ curve and is the energy-efficient detect choice for all current LoRa DtS-IoT deployments.

GEO satellites are outside the pass-arrival model because coverage is quasi-static; for devices inside the footprint, $f \approx 1$ and blind uplink is trivially optimal.³

NB-IoT NTN ($\kappa \approx 0.3$, $T_b \approx 20$ ms) provides an instructive contrast to LoRa. With these parameters, the closed-form gives $N^* \approx 546$ for $D = 0.01$: for constellations smaller than this, the detect overhead is justified, and satellite-detect saves energy; above it, a hypothetical blind/lightweight-access mode would be more energy-efficient. However, as discussed in Section III-E, the current IoT-NTN standard does not provide a blind-uplink option—the NPSS/NSSS/MIB/SIB31 acquisition chain is protocol-mandatory, so IoT-NTN devices always operate in the detect regime regardless of constellation size or energy optimality. The $N^* \approx 546$ result should be read as a model-derived design data point under the chipset-averaged κ approximation, indicating the constellation size beyond which a lightweight-access procedure (e.g. one that bypasses full

³The Poisson gap framework loses physical meaning when T_{orbit} and inter-pass gaps vanish; the blind/detect trade-off reduces to a different always-visible problem.

cell acquisition when ephemeris is recent) would become energy-favourable. Because κ here captures only the active-state RX/TX power split, a protocol-faithful accounting of the energy for NPSS/NSSS detection and SIB decoding could shift this threshold; quantifying the acquisition-procedure energy is left for future work.

VI. CONCLUSION

We have presented a first-order analytical design rule for event-driven blind versus satellite-detect uplink under intermittent LEO coverage. The closed-form crossover threshold $N^* = \log \alpha / \log(1 - f_1)$ depends on two physically intuitive parameters: the dimensionless beacon overhead $\alpha = \kappa_{\text{eff}} T_b / (2\tau_{\text{tx}})$ and the single-satellite coverage fraction f_1 . The threshold is valid in the dense-constellation regime ($\alpha \ll 1$, $f(N^*) \rightarrow 1$); for sparse Walker constellations with correlated pass geometry, propagated simulations are needed (Section IV-D, Table VI).

For LoRa-over-satellite parameters and $D = 1\%$, the threshold falls at $N^* \approx 220$ satellites. A lightweight ALOHA collision extension, cross-checked by Monte Carlo, shows that symmetric offered loads preserve this threshold, while asymmetric loads (higher blind contention) shift it upward or eliminate it entirely, confirming that the collision-free baseline is representative under balanced traffic but that differential contention requires fuller MAC-layer analysis [11]. For NB-IoT NTN under a chipset-averaged $\kappa \approx 0.3$ derived from public NB-IoT module datasheets [20]–[22], the threshold shifts to $N^* \approx 546$, identifying the constellation size beyond which a hypothetical lightweight-access mode would outperform the standard's mandatory acquisition chain; this result is a model-derived design data point under the active-state κ approximation, not a full acquisition-energy accounting.

Translated to deployment practice, the threshold provides a compact back-of-the-envelope rule for choosing between Class A-like and Class B-like operation on commodity satellite-IoT terminals across stationary, mobile, and vehicular use cases. Future work includes ephemeris-assisted scheduled access modelling for periodic reporting applications, mobility-aware variants accounting for predictable platform trajectories (e.g. rail or maritime), protocol-faithful NB-IoT NTN acquisition-energy accounting, multi-orbit and multi-shell constellations, and region-specific device density distributions.

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Juan A. Fraire is a researcher at INRIA (France) and a guest professor at CONICET-UNC (Argentina) and Saarland University (Germany). His core areas of interest are near-Earth and deep-space networking and informatics, with more than 120 publications in international journals and at leading conferences. Juan is co-chair of the Systems and Protocol Aspects for Circumstellar Environments (SPACE) Research Group at the IETF, the co-founder and chair of the Space-Terrestrial Inter-networking Workshop (STINT), and participates in

diverse joint projects with space agencies (e.g., NASA, ESA, CONAE) and companies in the space sector (e.g., D3TN, Skyloom).



Vincent Deslandes is Strategic Project Manager at Kinéis (Toulouse, France), where he oversees innovation projects and leads the technical roadmap of the Kinéis Gen 2 constellation, in addition to acting as telecom system architect for the company's end-to-end satellite IoT system. He has extensive experience as a system engineer and architect on LEO and non-terrestrial satellite programs, including Argos, Telesat LEO, Omnispace, and Astrocast (UNIT IoT protocol development), as well as on optical and quantum communications. His current

focus is on direct-to-satellite IoT architectures, with contributions to industry forums such as NTN Days on the role of IoT in 5G non-terrestrial networks.



Benoît Coeugnet is a Ph.D. candidate jointly affiliated with Kinéis (Toulouse, France), Inria (CITI Laboratory, INSA Lyon, France), and the IRIT laboratory (ENSEEIH, Toulouse INP, France) under a CIFRE industrial doctoral fellowship. His research focuses on direct-to-satellite IoT networks, with emphasis on analytical modelling and optimisation of access protocols, downlink scheduling, and energy-efficient device behaviour for LoRa-based satellite IoT. He contributes to the DORSAL-IoT project on next-generation LEO IoT systems and has presented

tutorials on direct-to-satellite IoT network simulation using FLoRaSat and OMNeT++.



Oana Iova is an Associate Professor in the Telecommunications Department at INSA Lyon and a member of the CITI laboratory and the Inria Agora research group. She has served as Deputy Director for International Relations at INSA Lyon since January 2024. She received her Ph.D. in Computer Science from the University of Strasbourg in 2014, with a thesis on standards optimisation and network lifetime maximisation for wireless sensor networks. Her research interests include low-power wide-area networks (LPWANs), MAC and routing protocols

for wireless sensor networks, IoT performance evaluation, and large-scale wireless testbeds. She has co-authored numerous publications on LoRa and IEEE 802.15.4-based systems and contributes to several national and European IoT projects.



André-Luc Beylot is a Full Professor in the Telecommunications and Networks Department of ENSEEIH, Toulouse INP, and a member of the IRIT laboratory (UMR 5505), which he has been head of at the ENSEEIH site since January 2011. He received his Engineering Degree from the Institut d'Informatique d'Entreprise (CNAM) in 1989 and his Ph.D. in Computer Science from the University of Paris 6 in 1993. From 2008 to 2011 he led the IRT team at IRIT, and from 2016 to 2020 he chaired the CNRS research network GDR RSD on Networks and

Distributed Systems. His research interests include performance evaluation of wireless, mobile, vehicular, and satellite networks, with a particular focus on access protocols and quality of service.